

POWER SWING BLOCK

Basic Concepts:

Power swing block is a protection function designed to prevent distance or impedance relay from tripping during power swing condition.

A power swing is a temporary oscillation in voltage & current in the system. Typically caused by large load change, line switching, or generator synchronizing without an actual fault.



Purpose:

- System oscillation occurs (voltage & current slowly).
- The distance relay impedance trajectory moves slowly in the R-X line.
- PSB detects this slow movement → consider it power swing.
- If sudden impedance occurs = indicate actual fault → PSB allows relay trip.
- Operate immediately (35-50 ms)

Relay Settings For Zone:

Zone-1	Zone-2	Zone-3
$X_{(rech)} = 34.062 \Omega$	$X_{(rech)} = 57.479 \Omega$	$X_{(rech)} = 85.155 \Omega$
$R(ph-g) = 20.636 \Omega$	$R(ph-g) = 34.822 \Omega$	$R(ph-g) = 51.588 \Omega$
$R(ph-ph) = 17.031 \Omega$	$R(ph-ph) = 28.739 \Omega$	$R(ph-ph) = 42.577 \Omega$

Equation:

$$\text{CT Ratio} = 4000/1 \text{ A}$$

$$\text{CT Multiplier} = \frac{4000}{1} = 4000\text{A}$$

$$\text{PT Ratio} = 400000/110 \text{ V}$$

$$\text{PT Multiplier} = \frac{400000}{110} = 3636.36 \text{ V}$$

$$Z_{\text{primary}} = \frac{V_{\text{primary}}}{I_{\text{primary}}} = \frac{V_{\text{sec}} * \text{PT multiply}}{I_{\text{sec}} * \text{CT multiply}}$$

$$= Z_{\text{sec}} * \frac{\text{PT multiply}}{\text{CT multiply}}$$

$$= Z_{\text{sec}} * \frac{3636}{4000} = Z_{\text{sec}} * 0.909$$

$$\frac{dz}{dt} = \text{Instantaneous rate of change of apparent impedance seen by the relay.}$$

$$\frac{dz}{dt} = \text{Small \& slow movement} \rightarrow \text{Power swing}$$

$$\frac{dz}{dt} = \text{Large \& Fast movement} \rightarrow \text{Actual fault}$$

$$\frac{dz}{dt} = \frac{Z(tn) - Z(tn-1)}{t1 - t2}$$

$$\text{Magnitude (Z)} = R + jX$$

$$|Z| = \sqrt{R^2 + X^2}$$

$$\begin{aligned} \text{Zone-1 (ph-g) Magnitude} &= \sqrt{(20.636)^2 + (34.062)^2} \\ &= 39.828 \, \Omega \end{aligned}$$

$$\text{Zone-2 (ph-g) Magnitude} = 67.205 \, \Omega$$

$$\text{Zone-3 (ph-g) Magnitude} = 99.563 \, \Omega$$

Example: Swing Condition

$$\frac{dz}{dt} = \frac{Z(tn) - Z(tn-1)}{t1 - t2}$$

$$\text{At } t1 = 0.00 \text{ s}$$

$$I_{\text{sec}} = 0.50 \text{ A}$$

$$V_{\text{sec}} = 105 \text{ v}$$

$$Z_{1\text{sec}} = \frac{105}{0.50} = 210 \Omega$$

$$Z_{\text{pri}} = 210 * 0.90 = 189 \Omega$$

$$\Delta Z = Z_2 - Z_1$$

$$= 176 - 189 = 13 \Omega$$

$$\frac{dz}{dt} = \frac{13}{0.050} = 250 \Omega/\text{s}$$

$$\text{Typically Swing Change} = 50 \Omega$$

$$\text{Swing Time} \sim 0.250 \text{ s}$$

$$\frac{dz}{dt} = \frac{50}{0.250} = 200 \Omega/\text{s}$$

Result : Its Swing . Not a fault

$$\text{At } t2 = 0.050 \text{ s}$$

$$I_{\text{sec}} = 0.50 \text{ A}$$

$$V_{\text{sec}} = 97 \text{ v}$$

$$Z_{2\text{sec}} = \frac{97}{0.50} = 194 \Omega$$

$$Z_{\text{pri}} = 194 * 0.90 = 176 \Omega$$

Example: Swing Condition

$$\text{At } t1 = 0.00 \text{ s}$$

$$I_{\text{sec}} = 1 \text{ A}$$

$$V_{\text{sec}} = 108 \text{ v}$$

$$Z_{1\text{sec}} = \frac{108}{1} = 108 \Omega$$

$$Z_{\text{pri}} = 108 * 0.90 = 97 \Omega$$

$$\Delta Z = Z_2 - Z_1$$

$$= 34 - 97 = 63 \Omega$$

$$\frac{dz}{dt} = \frac{63}{0.010} = 6300 \Omega/\text{s} \rightarrow \text{its actual fault}$$

$$\text{At } t2 = 0.010 \text{ s (10 ms)}$$

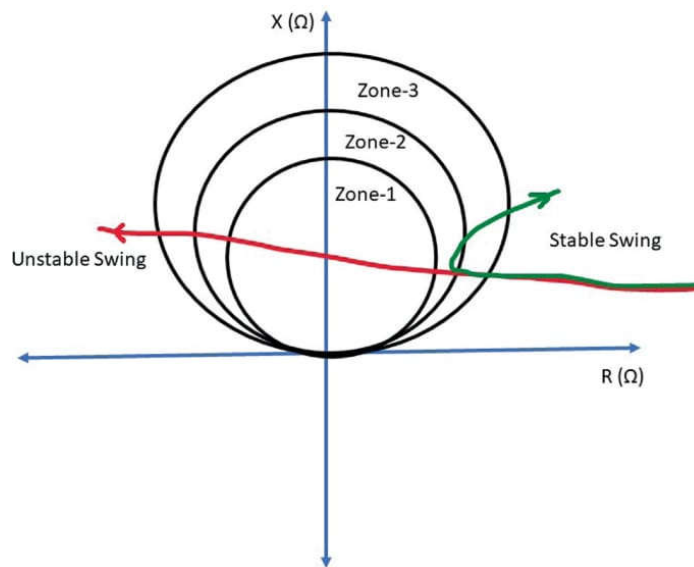
$$I_{\text{sec}} = 1 \text{ A}$$

$$V_{\text{sec}} = 38 \text{ v}$$

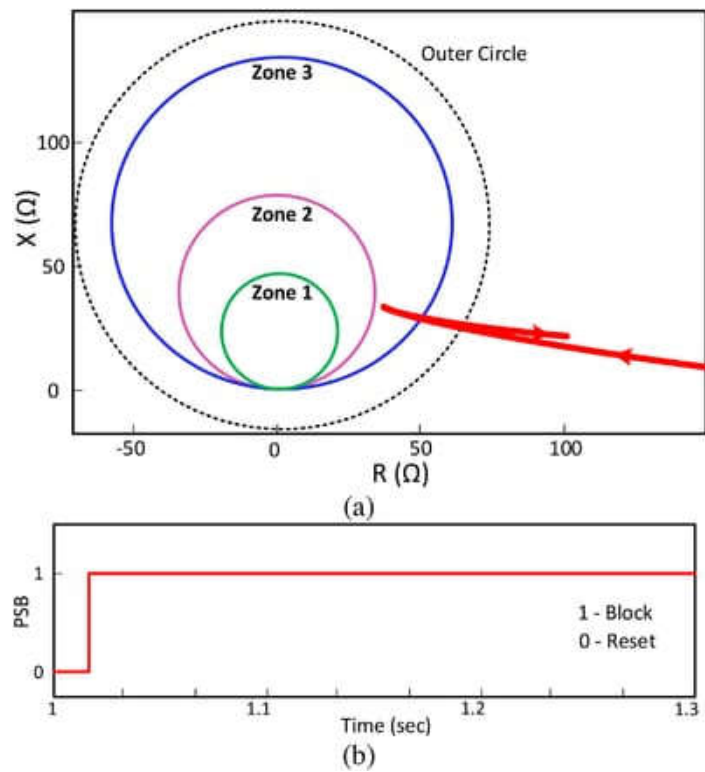
$$Z_{2\text{sec}} = \frac{38}{1} = 40 \Omega$$

$$Z_{\text{pri}} = 38 * 0.90 = 34 \Omega$$

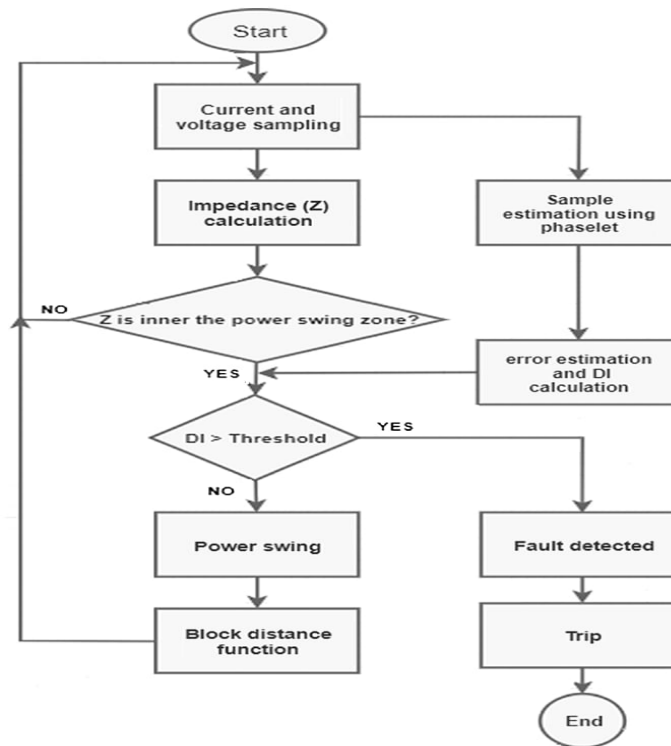
Stable swing Scenario:



Unstable swing Scenario:



Block Diagram:



Causes of Power Swing (Real System Scenario):

Causes	Cause Explanation	Example
1	Sudden Load Rejection	Large block of load suddenly trips (e.g., 300 MW rejection at Korerhat end). Generator mechanical power remains same → rotor accelerates → angle oscillation.
2	Loss of One Transmission Circuit	One 400 kV double-circuit line trips → power is transferred through the remaining line → sudden impedance and power flow change → oscillation starts.
3	Fault Near Bus with Delayed Clearing	Temporary voltage collapse due to fault; after clearing, voltage overshoots and starts oscillating.
4	Improper Generator Synchronization	Generator synchronized with grid having angle or frequency mismatch → transient torque → oscillation
5	Weak or Long Transmission Line	Long 400 kV line (e.g., Meghnaghat–Korerhat 140 km) has low damping; even small disturbances cause low-frequency oscillations.

Relay Behavior During Power Swing:

Parameter	During Fault	During Power Swing
Voltage (V)	Sudden deep drop (near zero)	Oscillates slowly (low–high–low)
Current (I)	Sudden high rise	Oscillates moderately with voltage
Impedance (Z)	Jumps directly inside Zone-1	Moves gradually from Zone-3 → Zone-2 → Zone-1 → Zone-2 (back)
dz/dt	Very high ($>2000 \Omega/s$)	Slow ($<200 \Omega/s$ typical)
Relay Trip Logic	Trips instantaneously (21/67 elements)	PSB (Power Swing Block) activates → blocks Zone elements temporarily
Duration	Short (<100 ms)	Longer (0.5–5 seconds typical)

Summary:

Power Swing is an oscillatory phenomenon in voltage, current, and impedance due to generator–system imbalance.

The relay detects the slow movement of impedance and activates PSB (Power Swing Blocking) to prevent unwanted tripping during stable swings.